LSM 6DSO `IMU 1` | `SPI 2`

- Registers

Addr (Hex)	Register Name	Data
01	FUNC_CFG_ACCESS	
0D	INT1_CTRL	`0b11`
0F	WHO_AM_I	
10	CTRL1_XL	`0b10000100`
11	CTRL2_G	`0b10001100`
12	CTRL3_C	`0b01000100`
14	CTRL5_C	
17	CTRL8_XL	`0b0`
19	CTRL10_C	`0b100000`
1E	STATUS_REG	
20	OUT_TEMP_L	TEMP_DATA[7:0]
21	OUT_TEMP_H	TEMP_DATA[15:8]
22	OUTX_L_G	GYRO_DATA_X[7:0]
23	OUTX_H_G	GYRO_DATA_X[15:8]
24	OUTY_L_G	GYRO_DATA_Y[7:0]
25	OUTY_H_G	GYRO_DATA_Y[15:8]
26	OUTZ_L_G	GYRO_DATA_Z[7:0]
27	OUTZ_H_G	GYRO_DATA_Z[15:8]
26	OUTX_L_A	ACCEL_DATA_X[7:0]
29	OUTX_H_A	ACCEL_DATA_X[15:8]
2A	OUTY_L_A	ACCEL_DATA_Y[7:0]
2B	OUTY_H_A	ACCEL_DATA_Y[15:8]
2C	OUTZ_L_A	ACCEL_DATA_Z[7:0]
2D	OUTZ_H_A	ACCEL_DATA_Z[15:8]
40	TIMESTAMP0	TIMESTAMP_DATA[31:24]
41	TIMESTAMP1	TIMESTAMP_DATA[23:16]

Addr (Hex)	Register Name	Data
42	TIMESTAMP2	TIMESTAMP_DATA[15:8]
43	TIMESTAMP3	TIMESTAMP_DATA[7:0]
56	TAP_CFG0	`0b01000001`

- Characteristics
  - Gyroscope
    - Gyroscope range: 2000 dps
    - Sensitivity for 2k dps: 70 mdps/LSB
    - Initial zero rate output tolerance: +- 1 dps
    - ODR: 1666 Hz
  - Accelerometer
    - Accelerometer range: 16 g
    - Sensitivity for 16 g: 0.488 mg/LSB
    - Initial zero rate output tolerance: +- 20 mg
    - ODR: 1666 Hz
  - Temperature
    - Sensitivity: 256 LSB/C
- SPI
  - Pull CS to activate: LOW
  - Data delivered MSB first LSB last
  - Data transitioned on falling edge of SCLK
- Self test at boot for later version
- Interrupts
  - INT fire pull: HIGH
  - Use INT 1
  - INT fired when DATA READY, cleared when INT STATUS bit READ
- Time stamp of reading stored in register, resolution of `25 us`
- Power mode: 6-axis with independant ODR
- Block data update